Meta analysis

Code

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▶ Load packages and setup

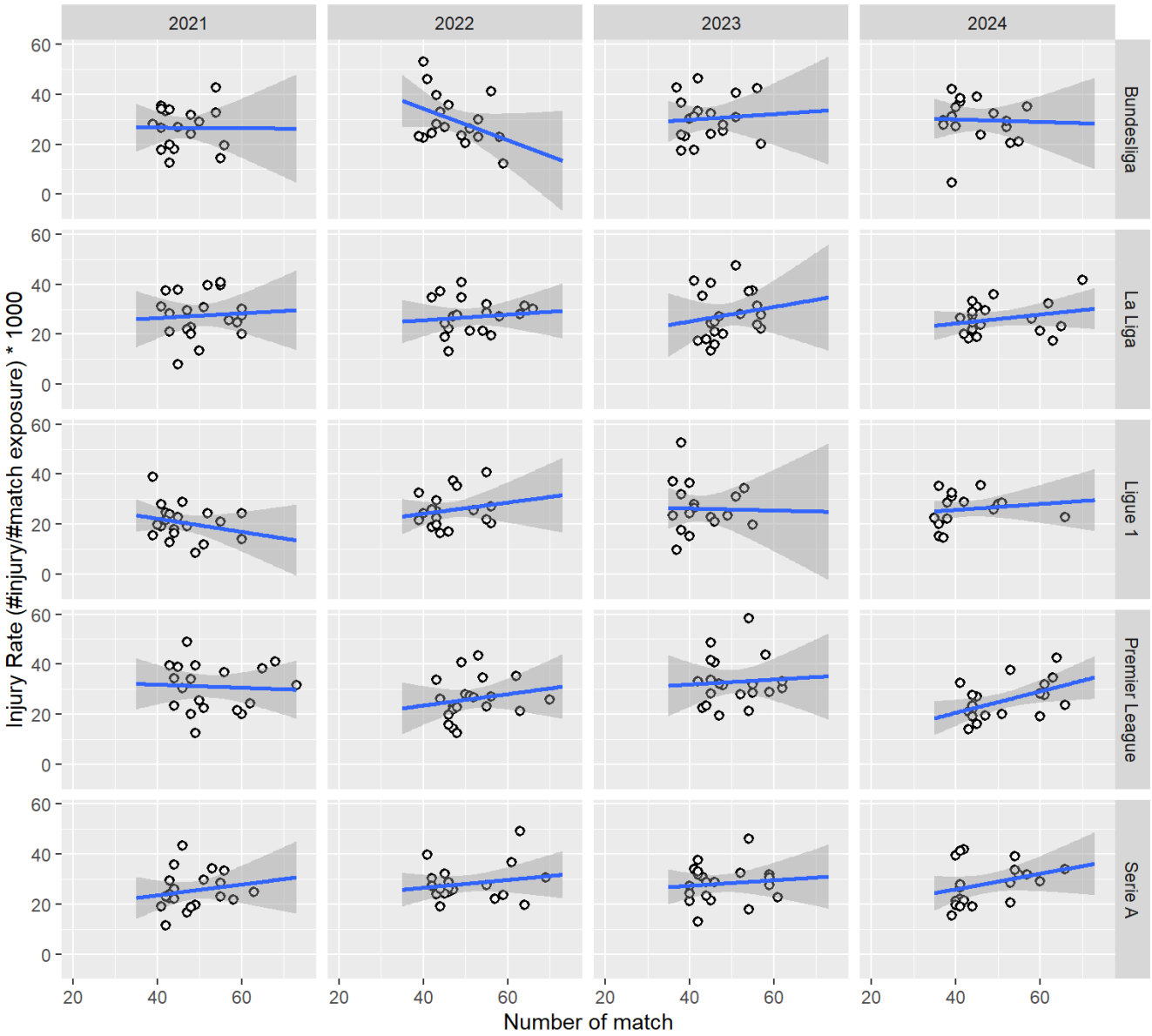
▶ Load dataset

```
Rows: 1,560
Columns: 10
$ country           <chr> "Germany", "Germany", "Germany", "Germany", "Germany…
$ league            <chr> "Bundesliga", "Bundesliga", "Bundesliga", "Bundeslig…
$ league_tag        <chr> "cp_9", "cp_9", "cp_9", "cp_9", "cp_9", "cp_9", "cp_…
$ team              <chr> "Eintracht Frankfurt", "FC St. Pauli von 1910", "Bor…
$ team_tag          <chr> "t_979", "t_1018", "t_971", "t_973", "t_972", "t_964…
$ season            <dbl> 2010, 2010, 2010, 2010, 2010, 2010, 2010, 2010, 2010…
$ n_match           <int> 37, 33, 40, 39, 36, 45, 45, 44, 38, 49, 55, 36, 37, …
$ n_injury          <int> 0, 2, 6, 0, 1, 2, 3, 0, 1, 2, 1, 0, 2, 1, 2, 1, 4, 1…
$ match_exposure    <dbl> 610.5, 544.5, 660.0, 643.5, 594.0, 742.5, 742.5, 726…
$ match_injury_rate <dbl> 0.000000, 3.673095, 9.090909, 0.000000, 1.683502, 2.…
```

Injury Rate (2019–2024)

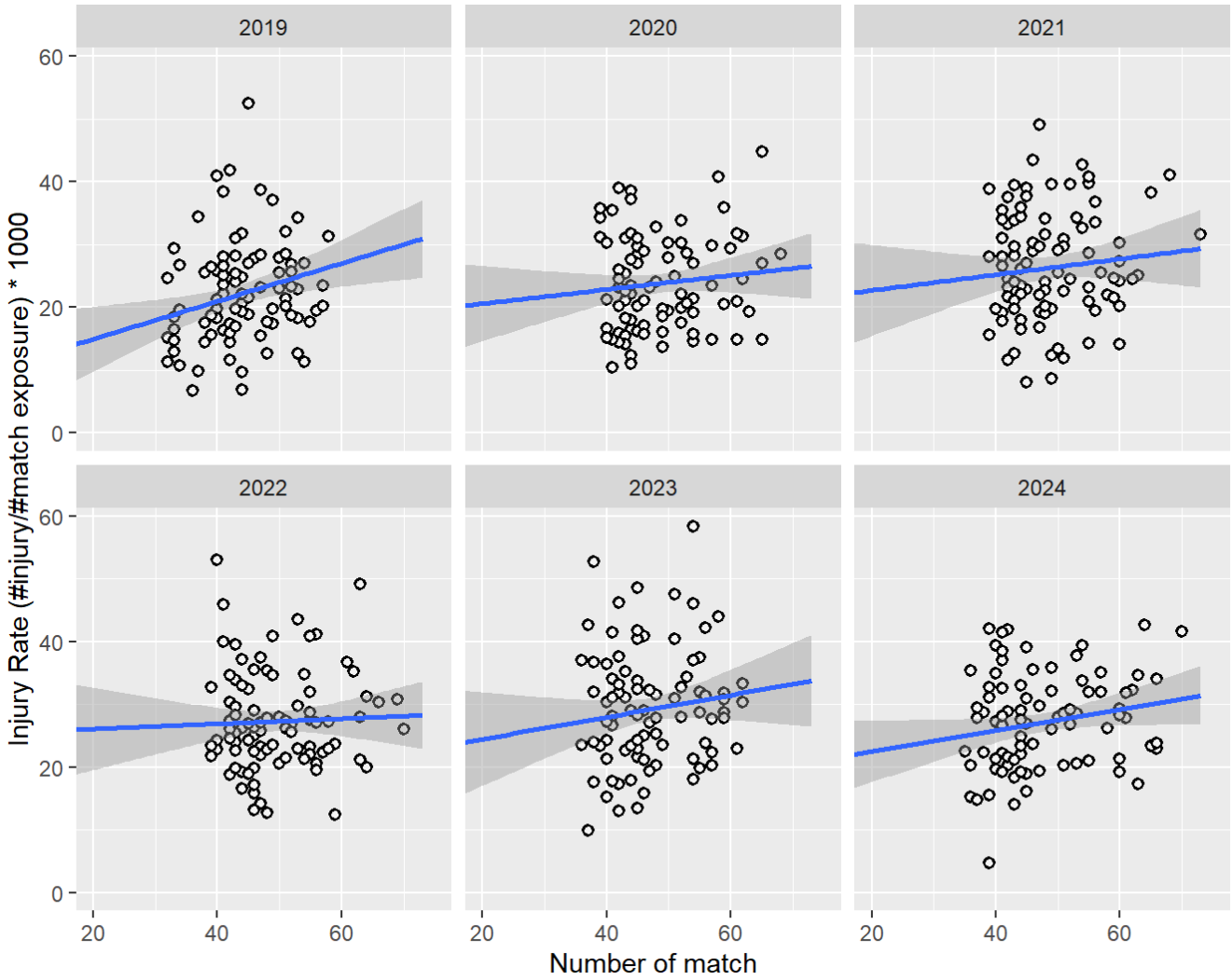
By league and season

▶ Plot match injury rate by league and season



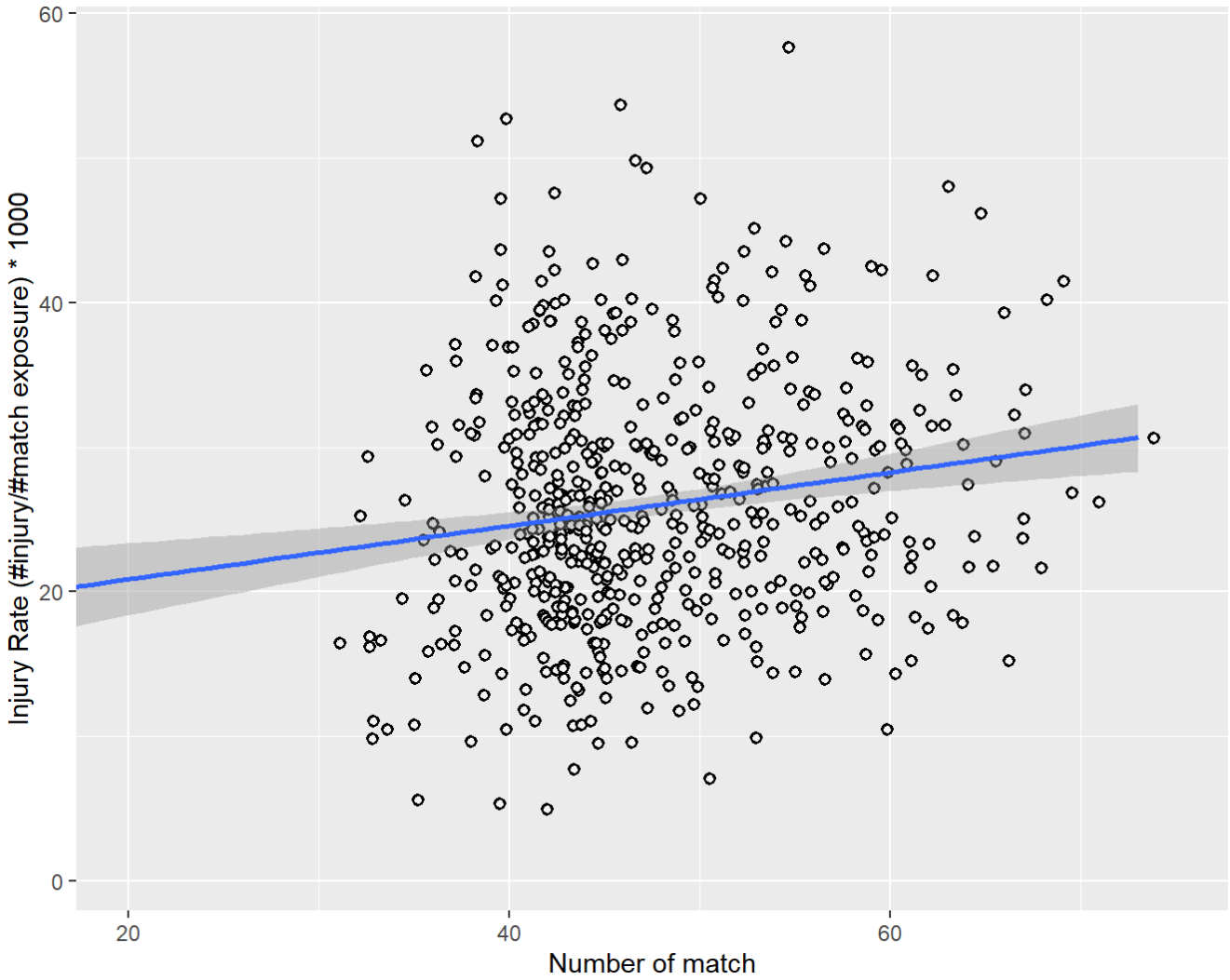
By season

▶ Plot match injury rate by season



Overall

▶ Plot match injury rate by season



Modelling

Joint Analysis

▶ Generalized mix-effect model - league:team as random factor

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + league + (1 | league:team)
   Data: mutate(match_injury, season_z = (season - mean(season))/sd(season),
    match_played_z = (n_match - mean(n_match))/sd(n_match))
 Offset: log(match_exposure)

      AIC       BIC    logLik -2*log(L)  df.resid
   9484.1    9526.9   -4734.0    9468.1      1552

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.4836 -0.9776 -0.1972  0.7968  5.5785
```

```
Random effects:
 Groups      Name        Variance Std.Dev.
 league:team (Intercept) 0.03377  0.1838
Number of obs: 1560, groups:  league:team, 186

Fixed effects:
                      Estimate Std. Error  z value Pr(>|z|)
(Intercept)          -4.187474   0.040075 -104.492  < 2e-16 ***
match_played_z        0.108310   0.008842   12.250  < 2e-16 ***
season_z              0.590194   0.008788   67.155  < 2e-16 ***
leagueLa Liga        -0.189128   0.055080   -3.434 0.000595 ***
leagueLigue 1        -0.258907   0.055330   -4.679 2.88e-06 ***
leaguePremier League -0.028789   0.053680   -0.536 0.591746
leagueSerie A        -0.051122   0.054246   -0.942 0.345977
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mtch__ sesn_z legLLg legLg1 lgPrmL
mtch_plyd_z  0.038
season_z    -0.073  0.198
leagueLaLig -0.716 -0.062 -0.008
leagueLigu1 -0.707  0.013  0.023  0.517
leaguPrmrLg -0.738 -0.076 -0.005  0.537  0.529
leagueSeriA -0.727 -0.014 -0.005  0.528  0.524  0.542
```

Separate model for each league

▶ Generalized mix-effect model by league - team as random factor

▶ Summary of league-specific models

Bundesliga

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + (1 | team)
   Data: .x
 Offset: log(match_exposure)

      AIC       BIC    logLik -2*log(L)  df.resid
   1864.1    1878.7    -928.0    1856.1       284

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.3135 -1.1645 -0.1465  0.9971  3.9583
```

```
Random effects:
 Groups Name        Variance Std.Dev.
 team   (Intercept) 0.03225  0.1796
Number of obs: 288, groups:  team, 32

Fixed effects:
               Estimate Std. Error  z value Pr(>|z|)
(Intercept)    -4.17963    0.04106 -101.787  < 2e-16 ***
match_played_z  0.12987    0.02200    5.904 3.54e-09 ***
season_z        0.58412    0.02022   28.890  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mtch__
mtch_plyd_z  0.084
season_z    -0.174  0.211
```

Ligue 1

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + (1 | team)
   Data: .x
 Offset: log(match_exposure)

      AIC       BIC    logLik -2*log(L)  df.resid
   1930.3    1945.4    -961.1    1922.3       316

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.3536 -0.9984 -0.1949  0.6639  5.7814

Random effects:
 Groups Name        Variance Std.Dev.
 team   (Intercept) 0.03913  0.1978
Number of obs: 320, groups:  team, 35

Fixed effects:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)    -4.39511    0.04214 -104.31  < 2e-16 ***
match_played_z  0.09453    0.01828    5.17 2.34e-07 ***
season_z        0.63137    0.01957   32.26  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
```

```
            (Intr) mtch__
mtch_plyd_z -0.075
season_z    -0.189  0.228
```

Premier League

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + (1 | team)
   Data: .x
 Offset: log(match_exposure)

      AIC       BIC    logLik -2*log(L)  df.resid
     1718      1733      -855      1710       310

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.3028 -0.8837 -0.2451  0.6767  3.9901

Random effects:
 Groups Name        Variance Std.Dev.
 team   (Intercept) 0.01689  0.1299
Number of obs: 314, groups:  team, 37

Fixed effects:
               Estimate Std. Error  z value Pr(>|z|)
(Intercept)    -4.46374    0.03433 -130.031  < 2e-16 ***
match_played_z  0.15404    0.02257    6.826 8.72e-12 ***
season_z        0.71986    0.02340   30.763  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mtch__
mtch_plyd_z  0.106
season_z    -0.180  0.344
```

Serie A

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + (1 | team)
   Data: .x
 Offset: log(match_exposure)
```

```
      AIC       BIC    logLik -2*log(L)  df.resid 
   2109.8    2124.9   -1050.9    2101.8       316 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.7088 -1.1177 -0.1309  0.9444  4.9699 

Random effects:
 Groups Name        Variance Std.Dev.
 team   (Intercept) 0.0492   0.2218  
Number of obs: 320, groups:  team, 41

Fixed effects:
               Estimate Std. Error z value Pr(>|z|)    
(Intercept)    -4.19373    0.04275 -98.110  < 2e-16 ***
match_played_z  0.10285    0.01836   5.601 2.13e-08 ***
season_z        0.49357    0.01809  27.290  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mtch__
mtch_plyd_z -0.104       
season_z    -0.077  0.147
```

La Liga

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: n_injury ~ match_played_z + season_z + (1 | team)
   Data: .x
 Offset: log(match_exposure)

      AIC       BIC    logLik -2*log(L)  df.resid 
   1818.3    1833.3    -905.1    1810.3       314 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.3026 -0.8290 -0.0729  0.6685  3.5975 

Random effects:
 Groups Name        Variance Std.Dev.
 team   (Intercept) 0.02586  0.1608  
Number of obs: 318, groups:  team, 41

Fixed effects:
               Estimate Std. Error  z value Pr(>|z|)
```

```
(Intercept)    -4.22886    0.03525 -119.973  < 2e-16 ***
match_played_z  0.10268    0.01931    5.317 1.05e-07 ***
season_z        0.57774    0.01873   30.853  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) mtch__
mtch_plyd_z  0.041
season_z    -0.221  0.091
```

Meta Analysis

▶ Gather coefficients from each league specific model

Fixed effect model

Summary

▶ Meta analysis using fixed-effect method

Intercept

```

Fixed-Effects Model (k = 5)

  logLik  deviance       AIC       BIC      AICc
-11.7067   46.6822   25.4134   25.0228   26.7467

I^2 (total heterogeneity / total variability):   91.43%
H^2 (total variability / sampling variability):  11.67

Test for Heterogeneity:
Q(df = 4) = 46.6822, p-val < .0001

Model Results:

estimate      se       zval    pval    ci.lb    ci.ub
 -4.3017  0.0173  -249.2399  <.0001  -4.3355  -4.2679  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

match_played_z

```
Fixed-Effects Model (k = 5)

  logLik  deviance       AIC       BIC      AICc
 12.2060    5.5076  -22.4120  -22.8025  -21.0786

I^2 (total heterogeneity / total variability):   27.37%
H^2 (total variability / sampling variability):  1.38

Test for Heterogeneity:
Q(df = 4) = 5.5076, p-val = 0.2391

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.1132  0.0089  12.7405  <.0001  0.0958  0.1306  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

season_z

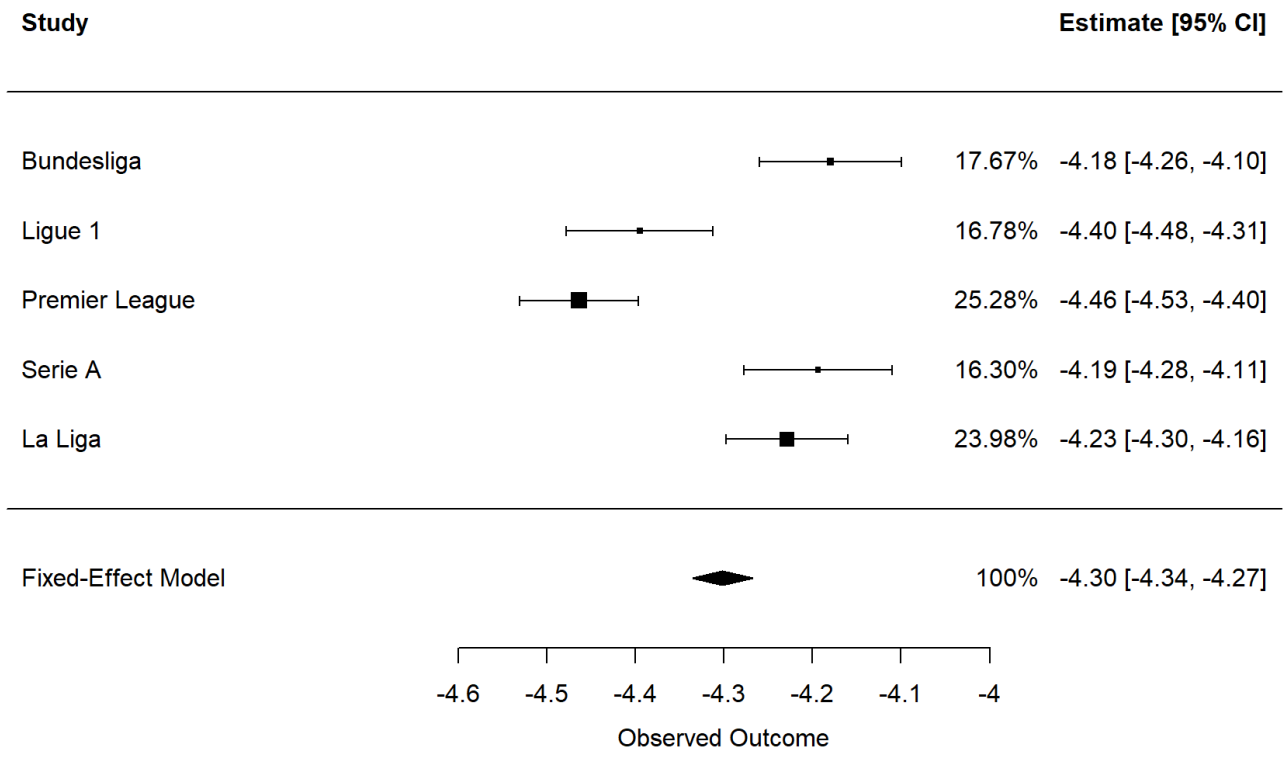
```
Fixed-Effects Model (k = 5)

  logLik  deviance       AIC       BIC      AICc
-17.1157   64.2028   36.2314   35.8408   37.5647

I^2 (total heterogeneity / total variability):   93.77%
H^2 (total variability / sampling variability):  16.05

Test for Heterogeneity:
Q(df = 4) = 64.2028, p-val < .0001

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.5901  0.0088  66.7407  <.0001  0.5728  0.6074  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

▶ Meta analysis using fixed-effect method

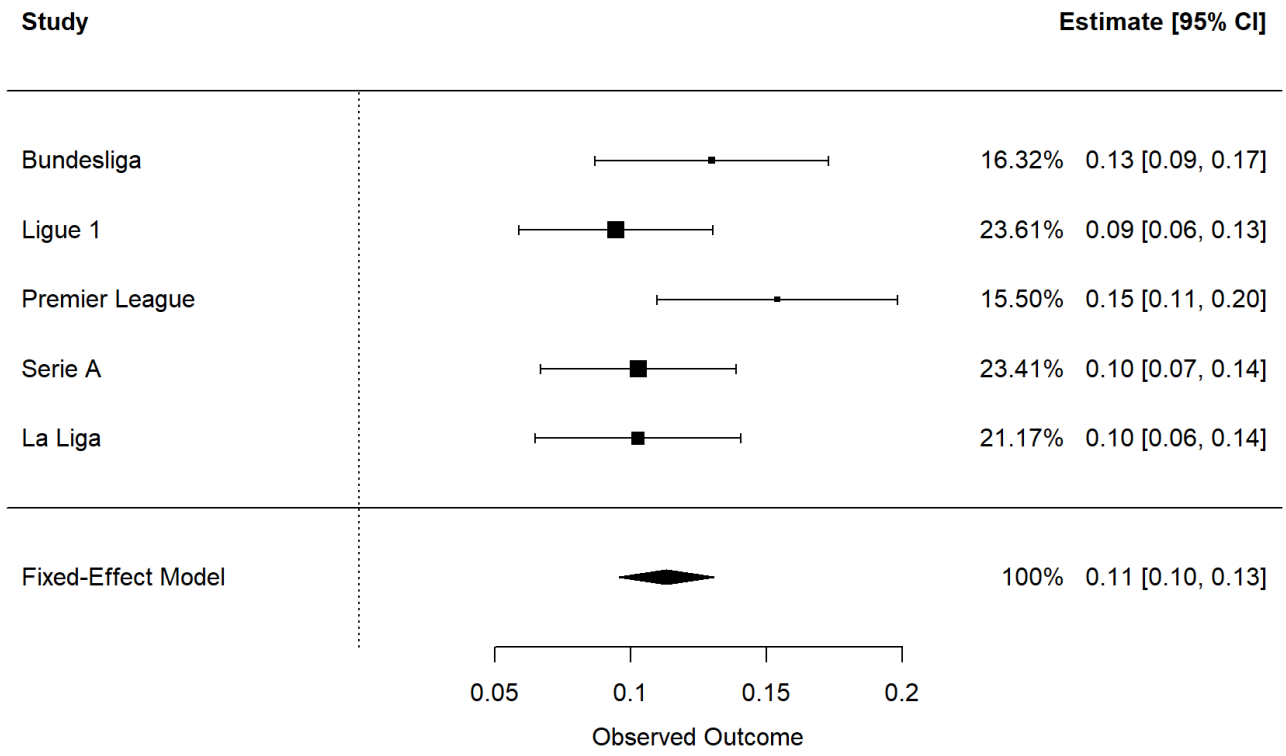
Plot

▶ Estimates from meta analysis using random-effect

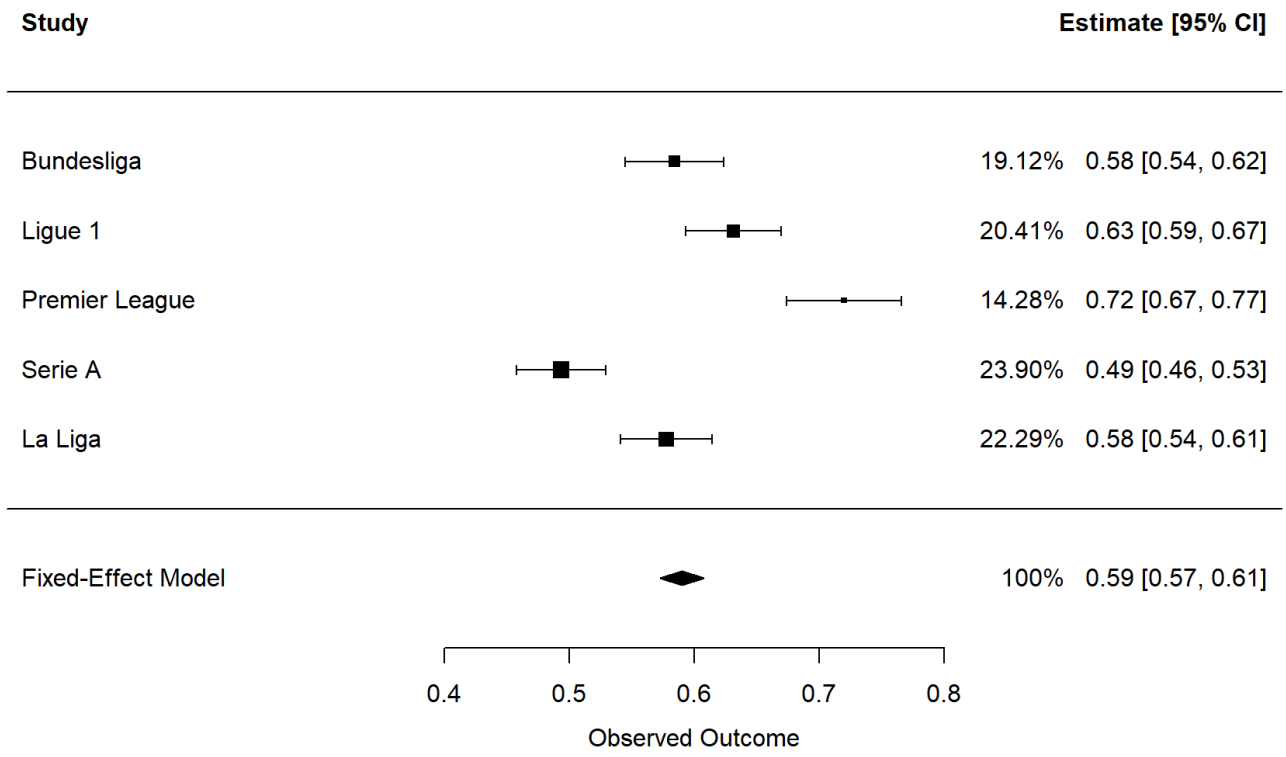
Intercept



match_played_z



season_z



▶ Estimates from meta analysis using random-effect

Random effect model

Summary

▶ Meta analysis using random-effect method

Intercept

```
Random-Effects Model (k = 5; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
  2.5121   -5.0241   -1.0241   -2.2515   10.9759

tau^2 (estimated amount of total heterogeneity): 0.0152 (SE = 0.0118)
tau (square root of estimated tau^2 value):      0.1233
I^2 (total heterogeneity / total variability):   91.01%
H^2 (total variability / sampling variability):  11.12

Test for Heterogeneity:
Q(df = 4) = 46.6822, p-val < .0001
```

```
Model Results:

estimate      se      zval    pval    ci.lb    ci.ub
 -4.2930  0.0579  -74.1743  <.0001  -4.4064  -4.1795  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

match_played_z

```
Random-Effects Model (k = 5; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
  9.2678  -18.5356  -14.5356  -15.7630   -2.5356

tau^2 (estimated amount of total heterogeneity): 0.0001 (SE = 0.0004)
tau (square root of estimated tau^2 value):      0.0110
I^2 (total heterogeneity / total variability):   23.26%
H^2 (total variability / sampling variability):  1.30

Test for Heterogeneity:
Q(df = 4) = 5.5076, p-val = 0.2391

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.1140  0.0102  11.1976  <.0001  0.0940  0.1339  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

season_z

```
Random-Effects Model (k = 5; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
  4.3017   -8.6034   -4.6034   -5.8308    7.3966

tau^2 (estimated amount of total heterogeneity): 0.0064 (SE = 0.0048)
tau (square root of estimated tau^2 value):      0.0798
I^2 (total heterogeneity / total variability):   94.18%
H^2 (total variability / sampling variability):  17.19

Test for Heterogeneity:
Q(df = 4) = 64.2028, p-val < .0001
```

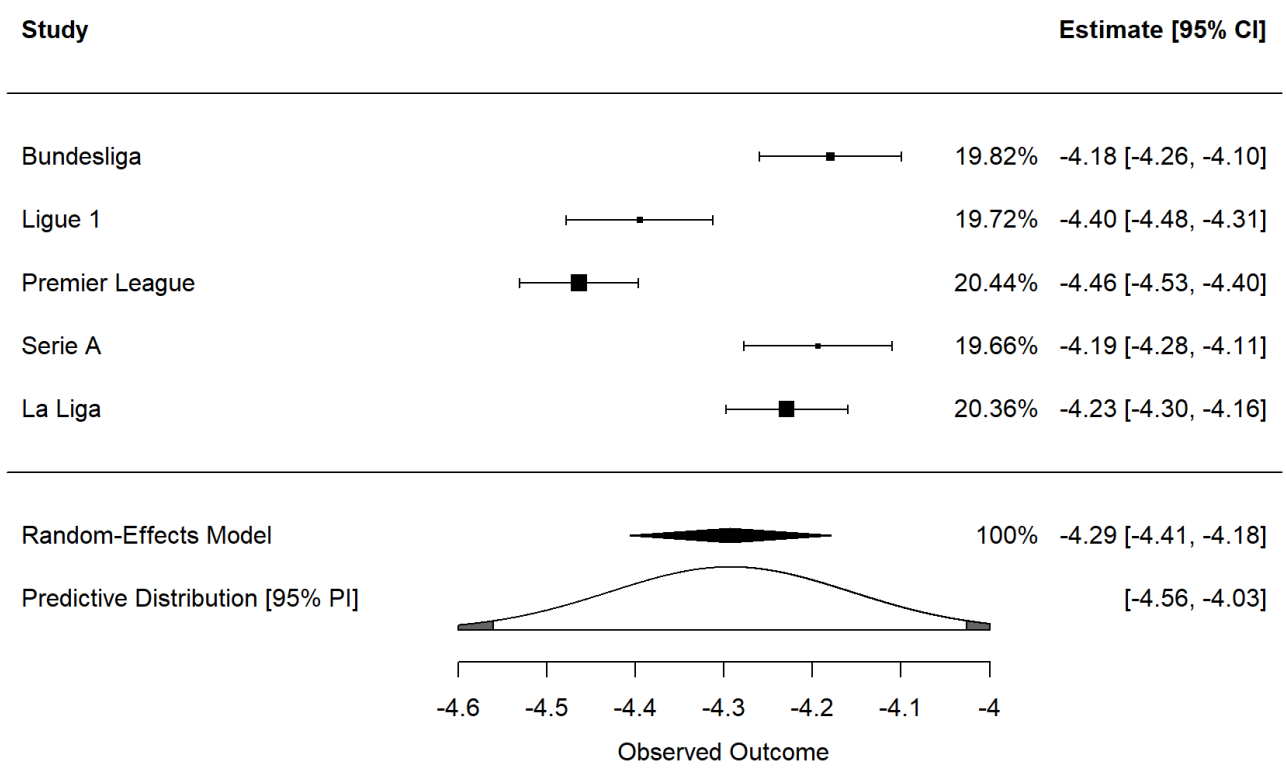
```
Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.6006  0.0368  16.3152  <.0001  0.5284  0.6727  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

▶ Meta analysis using random-effect method

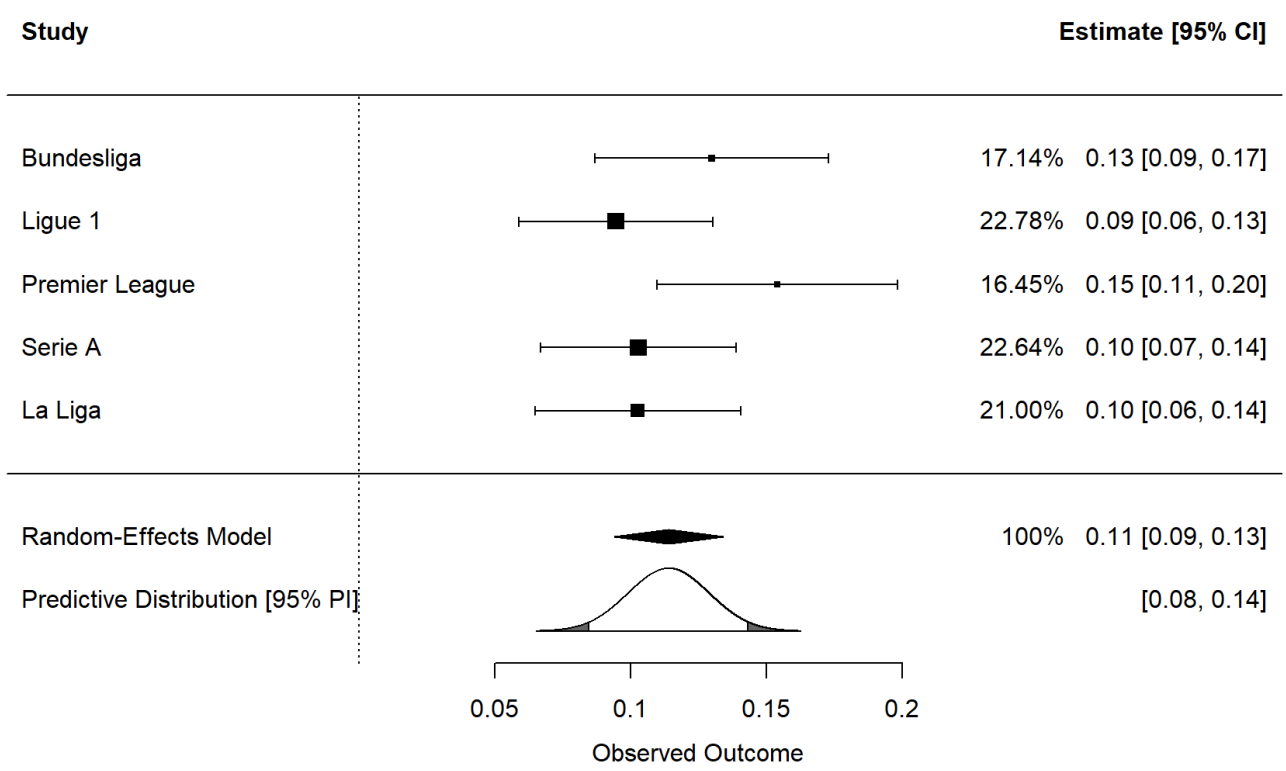
Plot

▶ Estimates from meta analysis using random-effect

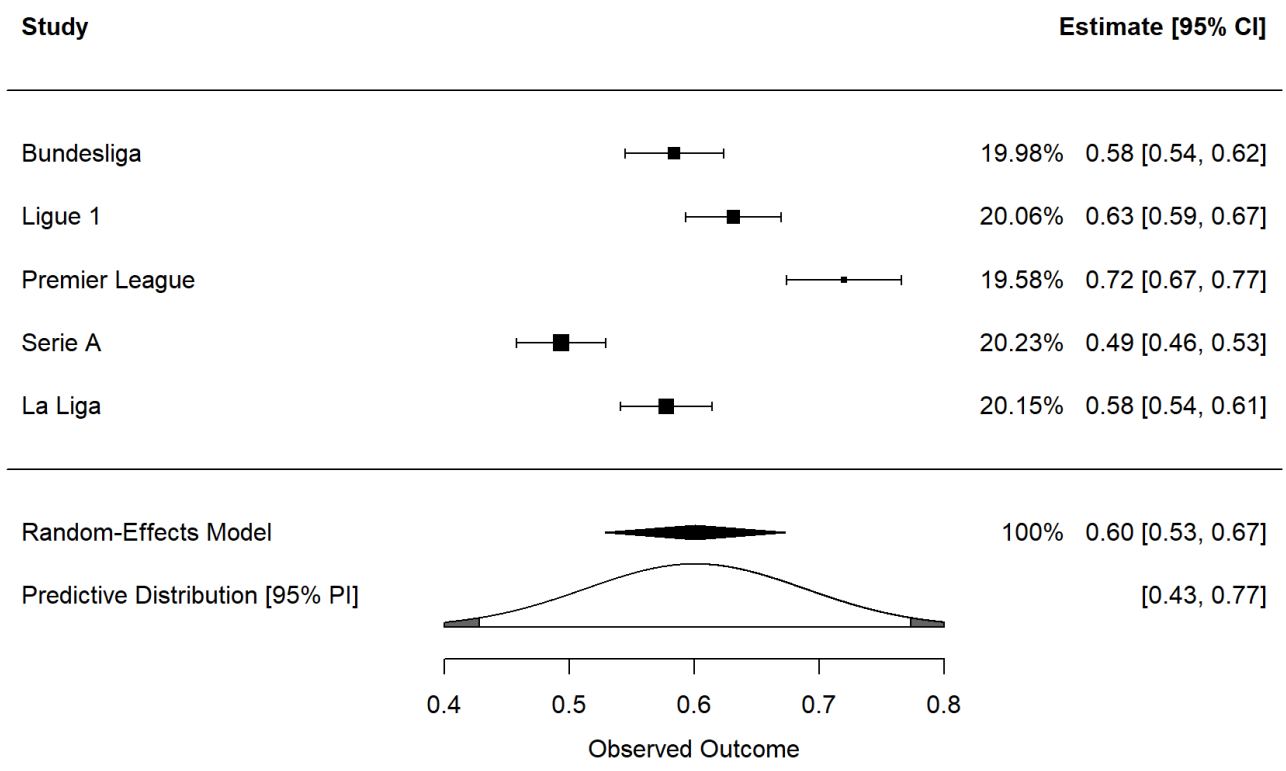
Intercept



match_played_z



season_z



▶ Estimates from meta analysis using random-effect

BLUPs

▶ Best Linear Unbiased Prediction (BLUP)

Intercept

BLUP Estimates

```
                  pred     se   pi.lb   pi.ub
Bundesliga     -4.1909 0.0394 -4.2681 -4.1137
Ligue 1        -4.3844 0.0403 -4.4635 -4.3054
Premier League -4.4515 0.0333 -4.5168 -4.3861
Serie A        -4.2044 0.0409 -4.2845 -4.1243
La Liga        -4.2337 0.0342 -4.3007 -4.1667
```

Pooled Estimates

```
    pred     se   ci.lb   ci.ub   pi.lb   pi.ub
 -4.2930 0.0579 -4.4064 -4.1795 -4.5600 -4.0260
```

match_played_z

BLUP Estimates

```
                 pred     se  pi.lb  pi.ub
Bundesliga     0.1171 0.0128 0.0921 0.1422
Ligue 1        0.1088 0.0120 0.0853 0.1324
Premier League 0.1216 0.0129 0.0964 0.1468
Serie A        0.1110 0.0120 0.0874 0.1346
La Liga        0.1112 0.0123 0.0872 0.1352
```

Pooled Estimates

```
   pred     se  ci.lb  ci.ub  pi.lb  pi.ub
 0.1140 0.0102 0.0940 0.1339 0.0846 0.1433
```

season_z

BLUP Estimates

```
                 pred     se  pi.lb  pi.ub
Bundesliga     0.5851 0.0197 0.5465 0.6238
Ligue 1        0.6296 0.0191 0.5921 0.6671
```

```
Premier League 0.7104 0.0226 0.6660 0.7548
Serie A        0.4988 0.0177 0.4640 0.5335
La Liga        0.5789 0.0183 0.5430 0.6149
```

Pooled Estimates

```
   pred     se  ci.lb  ci.ub  pi.lb  pi.ub
 0.6006 0.0368 0.5284 0.6727 0.4283 0.7729
```

▶ Best Linear Unbiased Prediction (BLUP)

AI Assisted Report

Summarizing meta-analysis results

I need to summarize and analyze the meta-analysis results in a structured way, using both math and plain language. I'll break it down into sections like Introduction, Data & Outcome, Primary Joint Model, League-Specific Models, Meta-analysis of Coefficients, and Practical Interpretation. I'll also touch on limitations and conclusions. I need to explain concepts like the log link, Poisson distribution, and how to compute the Injury Rate Ratio (IRR) using exponentiation, like exp(0.114). I'll keep it concise but complete. # Injury meta analysis report across the top 5 European leagues

You've already done a lot of the heavy lifting with the models—what I'll do here is turn those results into a coherent story, moving between the math and what it means in football terms.

---

1. Data and outcome

The dataset contains:

- **Leagues:** Bundesliga, La Liga, Ligue 1, Premier League, Serie A
- **Units:** Team–season observations (1,560 rows, 186 teams)
- **Outcome:**
  - `n_injury` = number of match injuries per team per season
  - **Exposure:** `match_exposure` (match minutes or match-hours), used as an **offset**
- **Key predictors:**
  - `n_match` (standardized to `match_played_z`): number of matches played
  - `season` (standardized to `season_z`): time trend (2010–2024)
  - `league`: categorical factor
- **Model family:** Poisson with log link, i.e. modeling **injury counts** with **log(injury rate)** as the linear predictor.

In plain language: you're modeling **injury rates per unit of match exposure**, adjusting for how many matches a team plays and how late in the time period the season is, with teams nested in leagues.

---

2. Joint generalized linear mixed model

2.1 Model specification

Joint model:

$$\log(\lambda_{ij}) = \beta_0 + \beta_1 \text{match_played_z}_{ij} + \beta_2 \text{season_z}_{ij} + \beta_{\text{league}} + u_{\text{league:team}},$$

with

- λ_{ij} = expected injury rate for team j in league i per unit exposure
- Offset: $\log(\text{match_exposure})$
- Random intercept: $u_{\text{league:team}} \sim N(0, \sigma^2)$

2.2 Key fixed effects (joint model)

From the joint model:

- **Intercept:** $\hat{\beta}_0 = -4.1875$
- **Match played (z):** $\hat{\beta}_1 = 0.1083$
- **Season (z):** $\hat{\beta}_2 = 0.5902$
- **League effects (vs Bundesliga as reference):**
  - La Liga: -0.1891
  - Ligue 1: -0.2589
  - Premier League: -0.0288 (ns)
  - Serie A: -0.0511 (ns)

All p-values for `match_played_z` and `season_z` are < 0.001.

2.3 Interpretation in plain language

- **Baseline injury rate:**
  The intercept -4.19 is the log injury rate at:

  - average number of matches,
  - average season (centered),
  - in the reference league (Bundesliga),
  - for an "average" team.

  Converting to a rate:

$$\text{Rate}_0 = e^{-4.1875} \approx 0.0152$$

So roughly **15 injuries per 1,000 units of exposure** (depending on how exposure is scaled).

- **Effect of number of matches (`match_played_z`):**

$$\text{IRR}_{\text{match}} = e^{0.1083} \approx 1.11$$

A **1 SD increase in number of matches** is associated with about an **11% higher injury rate**, holding season and league constant.

- **Effect of season (`season_z`):**

$$\text{IRR}_{\text{season}} = e^{0.5902} \approx 1.80$$

A **1 SD increase in season (i.e., moving towards more recent years)** is associated with about an **80% higher injury rate**, after adjusting for matches and league. This is a very strong upward time trend.

- **League differences (vs Bundesliga):**
  - La Liga: $e^{-0.1891} \approx 0.83$ → about **17% lower injury rate** than Bundesliga, all else equal.
  - Ligue 1: $e^{-0.2589} \approx 0.77$ → about **23% lower injury rate**.
  - Premier League and Serie A: small, non-significant differences vs Bundesliga.

In words: **injury rates are increasing over time**, higher for teams that play more matches, and slightly lower in La Liga and Ligue 1 compared to Bundesliga, with Premier League and Serie A roughly similar to Bundesliga once you adjust for other factors.

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3. League-specific mixed models

You then fit **separate Poisson mixed models by league**, each with:

$$\log(\lambda_{jt}) = \alpha_0 + \alpha_1 \text{match_played_z}_{jt} + \alpha_2 \text{season_z}_{jt} + u_{\text{team}},$$

with random intercepts for teams.

3.1 Coefficients by league

Intercepts (log rates):

- Bundesliga: -4.18
- Ligue 1: -4.40
- Premier League: -4.46
- La Liga: -4.19

- Serie A: -4.23

These are all in a fairly narrow band, roughly -4.46 to -4.18.

Match played (z):

- Bundesliga: 0.1299
- Ligue 1: 0.0945
- Premier League: 0.1540
- La Liga: 0.1029
- Serie A: 0.1027

All positive, all highly significant.

Season (z):

- Bundesliga: 0.5841
- Ligue 1: 0.6314
- Premier League: 0.7199
- La Liga: 0.4936
- Serie A: 0.5777

Again, all positive and highly significant, but with noticeable variation across leagues.

3.2 Interpretation

- **Within each league**, more matches and later seasons are consistently associated with **higher injury rates**.
- The **magnitude** of the season effect differs:
  - Strongest in the **Premier League** (~0.72),
  - Weakest in **La Liga** (~0.49),
  - Others in between (~0.58–0.63).

So the **direction of effect is stable across leagues**, but the **strength** of the time trend and baseline level varies.

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4. Meta-analysis of league-specific coefficients

You then treat each league's coefficient as a "study" and perform meta-analyses (k = 5) for:

- Intercept
- `match_played_z`
- `season_z`

using both **fixed-effect** and **random-effects** models.

4.1 Intercept

Fixed-effect model

- Estimate: -4.3017
- SE: 0.0173
- 95% CI: $[-4.3355, -4.2679]$
- Heterogeneity:
  - $I^2 = 91.43\%$ → very high
  - Q-test p < 0.0001

Interpretation: the **average baseline log injury rate** across leagues is about -4.30, but there is **substantial between-league heterogeneity** in intercepts.

Random-effects model

- Estimate: -4.2930
- SE: 0.0579
- 95% CI: $[-4.4064, -4.1795]$
- Predictive interval: $[-4.56, -4.03]$
- $I^2 = 91.01\%$, $\tau \approx 0.123$

Converting the pooled intercept:

$$e^{-4.2930} \approx 0.0137$$

So roughly **14 injuries per 1,000 exposure units** on average, with meaningful variation across leagues.

Plain language:
The **average baseline injury rate** across the five leagues is similar, but the high I^2 tells us that **leagues differ more than we'd expect by chance**. The random-effects model acknowledges that and gives a wider interval and a predictive range for a "new" league.

4.2 `match_played_z`

Fixed-effect model

- Estimate: 0.1132
- SE: 0.0089
- 95% CI: $[0.0958, 0.1306]$
- $I^2 = 27.37\%$, Q p = 0.2391 (no strong evidence of heterogeneity)

Random-effects model

- Estimate: 0.1140

- SE: 0.0102
- 95% CI: $[0.0940, 0.1339]$
- Predictive interval: $[0.0846, 0.1433]$
- $I^2 = 23.26\%$, very low τ^2

Exponentiating:

$$\mathrm{IRR}_{\text{match, pooled}} = e^{0.1140} \approx 1.12$$

So a **1 SD increase in matches** is associated with about a **12% higher injury rate**, and this effect is **remarkably consistent across leagues**.

Plain language:
The relationship between **number of matches and injury risk** is **stable and robust** across all five leagues. More matches → more injuries, and the size of that effect is similar everywhere.

4.3 `season_z`

Fixed-effect model

- Estimate: 0.5901
- SE: 0.0088
- 95% CI: $[0.5728, 0.6074]$
- $I^2 = 93.77\%$, Q p < 0.0001 → very high heterogeneity

Random-effects model

- Estimate: 0.6006
- SE: 0.0368
- 95% CI: $[0.5284, 0.6727]$
- Predictive interval: $[0.4283, 0.7729]$
- $I^2 = 94.18\%$, $\tau \approx 0.080$

Exponentiating:

$$\mathrm{IRR}_{\text{season, pooled}} = e^{0.6006} \approx 1.82$$

So a **1 SD increase in season** (moving forward in time) is associated with about an **82% higher injury rate on average**, but the **strength of this trend varies a lot between leagues**.

Plain language:
Injury rates are **rising over time in all leagues**, but **not equally**. Some leagues (e.g., Premier League) show a stronger increase than others (e.g., La Liga). The random-effects model captures this by giving a wider interval and a predictive range for what you might see in another league.

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5. BLUPs and pooled predictions

The BLUPs (Best Linear Unbiased Predictions) give **league-specific predicted coefficients** under the random-effects meta-analytic framework.

5.1 Intercept BLUPs

- Range roughly from -4.45 (Premier League) to -4.20 (La Liga), with narrow prediction intervals per league.
- Pooled estimate: -4.2930 (as above).

Interpretation:
Leagues differ modestly in baseline injury rates, but all sit in a fairly tight band around the pooled value.

5.2 `match_played_z` BLUPs

- League-specific predictions: ~0.108–0.122
- Pooled: 0.1140, 95% CI $[0.0940, 0.1339]$

Interpretation:
The **effect of match load is extremely consistent**—every league's BLUP is close to the pooled estimate.

5.3 `season_z` BLUPs

- League-specific predictions:
  - La Liga: ~0.50
  - Bundesliga/Serie A: ~0.58
  - Ligue 1: ~0.63
  - Premier League: ~0.71
- Pooled: 0.6006, 95% CI $[0.5284, 0.6727]$

Interpretation:
All leagues show **positive time trends**, but the **Premier League sits at the high end**, La Liga at the low end, with others in between.

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6. Practical interpretation and implications

6.1 Core findings

- **Injury rates are increasing over time** (large, consistent positive `season_z` effect).
- **More matches** → **more injuries**, with a **stable ~10–15% increase per SD of match load** across all leagues.

- **Baseline injury levels differ modestly between leagues**, with La Liga and Ligue 1 slightly lower than Bundesliga, and Premier League/Serie A similar to Bundesliga.
- **Heterogeneity is high for intercepts and time trends**, but **low for the match-load effect**.

6.2 What this means in football terms

- **Workload management:**
  The match-load effect is robust: **adding more matches reliably increases injury risk**, regardless of league. This supports policies around squad rotation, fixture congestion management, and rest periods.

- **Temporal trend:**
  The strong positive time trend suggests that **modern football is becoming more injury-prone**—possibly due to higher intensity, tighter schedules, or changes in style of play. The fact that this trend is stronger in some leagues (e.g., Premier League) hints at **league-specific structural or tactical factors**.

- **Cross-league comparisons:**
  While leagues differ somewhat in baseline injury rates, the **mechanisms linking load and time to injuries are broadly similar**. This makes cross-league lessons and interventions more transferable, especially around match congestion.

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7. Limitations and considerations

- **Poisson assumption:**
  If there is overdispersion (variance > mean), a negative binomial model might be more appropriate. The mixed model and random effects help, but don't fully guarantee perfect fit.
- **Standardization:**
  Effects are in units of **standard deviations**, which is great for comparability but less intuitive for practitioners. Translating SDs into actual match counts per league would help communication.
- **Unmeasured confounders:**
  No direct adjustment for training load, travel, schedule density, or rule changes—these may partly drive the time trend.
- **Exposure scaling:**
  Interpretation in "per 1,000 exposure units" depends on how `match_exposure` is defined (minutes, hours, etc.). Making that explicit would sharpen the practical meaning.

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8. Short, plain-language summary

Across the top five European leagues, teams that **play more matches** and seasons in **more recent years** have **higher match injury rates**. The effect of **match load** is consistent across leagues, while the **increase over time** is strong everywhere but varies in size—strongest in the Premier League, weakest in La Liga. Baseline injury levels differ somewhat between leagues, but not dramatically. Overall, the meta-analysis paints a clear picture: **modern football is getting more injury-intensive, and fixture congestion is a key, stable risk factor across competitions.**

If you'd like, we can next translate these log-scale effects into **concrete example scenarios** (e.g., "what happens if a team plays 5 extra matches?") for a report or presentation.